

VINAY SHANKER

Bengaluru, Karnataka

vinaymuddhe@gmail.com | linkedin.com/in/vinay-shanker-135660291 | github.com/vinayshanker9 |
portfolio-vinay-shanker.vercel.app

PROFESSIONAL SUMMARY

Proactive beginner Full Stack Developer with hands-on experience in Spring Boot and React ecosystems, actively building and deploying robust applications. Contributed to a patent-pending AI platform and developed a personal portfolio, demonstrating proficiency in RESTful API design and PostgreSQL.

EDUCATION

Jain (Deemed-to-be University), Faculty of Engineering & Technology
B.Tech in Information Science & Engineering

Aug 2023 – Present
Bengaluru, Karnataka

RELEVANT COURSEWORK

- Data Structures & Algorithms • Database Management Systems • Web Technologies • Operating Systems
- Object-Oriented Programming • Computer Networks • Software Engineering • Artificial Intelligence

PROJECTS

Solo Tutor — AI-Powered Learning Platform | *Python, FastAPI, React, Supabase*

2024 – Present

- Patent Filing In Progress · 5 Student Co-Inventors
- Built a multi-modal RAG platform enabling hallucination-free Q&A from documents and YouTube URLs using persistent vector embeddings (pgvector) in Supabase.
- Engineered automated quiz generation and YouTube transcript analysis with Whisper-based audio fallback; named co-inventor on the patent application at Jain University.

Distributed Rate Limiter & API Gateway | *Java, Spring WebFlux, Redis, Docker*

2026

- Built a high-throughput API Gateway using Token Bucket rate-limiting with Spring WebFlux for non-blocking I/O.
- Used Redis with Lua scripts for atomic distributed state management; load-tested with JMeter.

Event-Driven E-Commerce Order Engine | *Spring Boot, Kafka, PostgreSQL, Docker*

2026

- Designed a fault-tolerant microservices architecture separating Order, Inventory, and Payment domains using the Saga Pattern.
- Orchestrated asynchronous event streams with Apache Kafka; implemented distributed tracing with Zipkin.

TECHNICAL SKILLS

Languages: Java, JavaScript, HTML, CSS, SQL

Frameworks & Libraries: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, REST APIs, JWT Authentication, React.js

Databases: PostgreSQL, MongoDB

AI: Spring AI, RAG, pgvector

Developer Tools: Git, GitHub, Maven, Gradle, Docker

ACHIEVEMENTS & EXTRACURRICULAR

- Named co-inventor on the Solo Tutor Invention Disclosure Form and patent application filed at Jain University — recognising contribution to the AI-powered quiz generation module.
- Self-learning Spring Boot and Spring AI ecosystem through project-based practice, building REST APIs and exploring AI integration for backend services.